Telco Customer Churn Overview

Total Customers	Churn Rate	Monthly Revenue at Risk	Average Tenure (Months)
7,043	26.5%	$139.1K	32.4

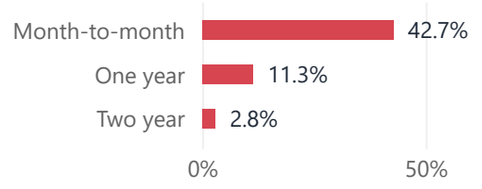
Contract: All

Internet Service: All

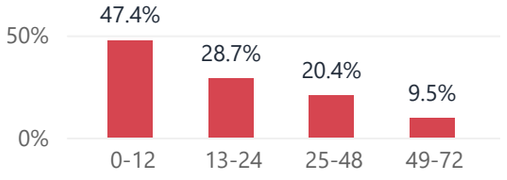
Tenure Band: All

Senior Citizen: All

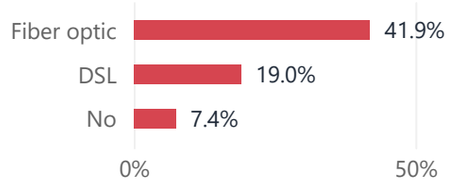
Churn Rate by Contract



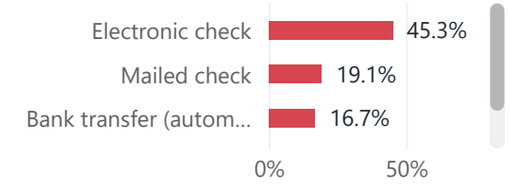
Churn Rate by Tenure Band



Churn Rate by Internet Service



Churn Rate by Payment Method



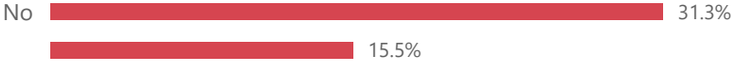
Churn Rate — Senior Citizen



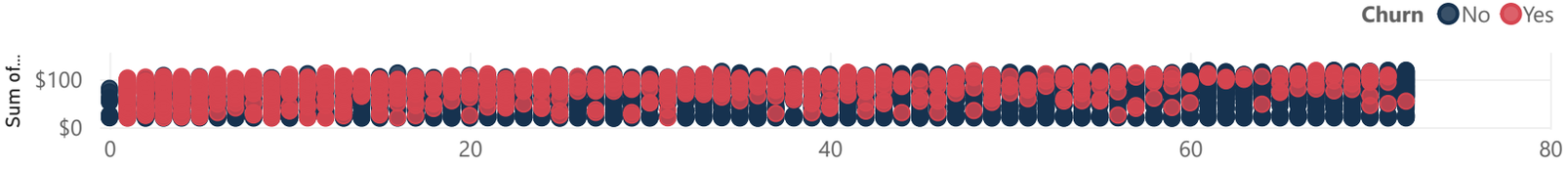
Churn Rate — Partner



Churn Rate — Dependents



Monthly Charges vs Tenure by Churn



Key Findings

- Overall churn is 26.5% (1,869 of 7,043); $139.1K in monthly revenue is at risk.
- Month-to-month churn is 42.7%, versus 2.8% for two-year contracts.
- Churn falls from 47.4% at 0–12 months to 9.5% at 49–72 months.
- Electronic check (45.3%) and fiber optic (41.9%) are the highest-risk payment and service groups.

Data Quality & Transformation Validation

PASS

Overall Data Quality

Row Integrity & Customer Key Checks

7,043 rows | 0 null customer IDs | 0 duplicate customer IDs | 0 null TotalCharges

The row count equals the distinct customer count, confirming one record per customer in this dataset snapshot.

Required-Field Null Checks

0 null tenure values | 0 null MonthlyCharges | 0 null TotalCharges | 0 null Churn labels

All dashboard-critical fields are complete after the documented Power Query transformations.

TotalCharges Fix Reconciliation

11 tenure-zero customers = 11 zero TotalCharges records

Reconciliation result: MATCH

Why TotalCharges blanks became zero

The source contained 11 blank TotalCharges values, all belonging to customers with tenure = 0. These represent new customers with no accumulated billing history, so the blanks were replaced with 0 rather than deleting the rows or estimating charges.

Verification: 11 tenure-zero customers = 11 zero TotalCharges records; post-fix TotalCharges nulls = 0.

Quality controls applied

- Confirmed the transformed table contains 7,043 rows and 7,043 distinct customer IDs.
- Checked customerID, tenure, MonthlyCharges, TotalCharges and Churn for missing values after transformation.
- Converted SeniorCitizen from 0/1 to No/Yes and created explicit churn, tenure-band and monthly-charge-band fields.
- **Reconciled the TotalCharges correction against tenure-zero customers. All documented checks pass.**

Scope note: these checks validate completeness, uniqueness and the known TotalCharges issue; they do not prove that every source value is operationally correct.